SREE RAMA ENGINEERING COLLEGE, TIRUPATI

(AUTONOMOUS)

MACHINE LEARNING APPLICATIONS IN WAVE ENERGY FORECASTING

PRESENTED BY BATCH:2

Team Members -1. P. SOWMYA - 224C1A3051
2. D. SREENATH - 224C1A3019
3. P. NIROSHA - 224C1A3052
4. K. ALEKHYA - 224C1A3033
5. C. S. NITHESH - 224C1A3014

Under The Guidance of
Mr. C Govardhan
Assistant Professor

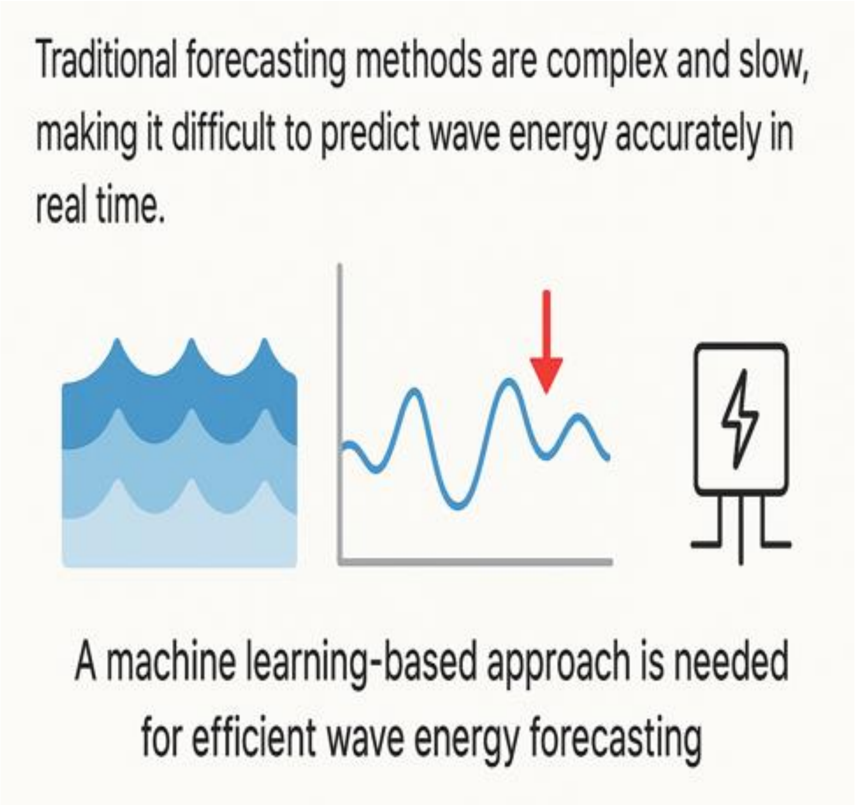
Contents

INTRODUCTION

- Wave energy is a clean and renewable source of energy from ocean waves
- It is used to generate electricity without pollution
- Accurate wave energy forecasting is important for efficient power generation
- Ocean waves are complex and change continuously
- Machine Learning helps predict wave energy and wave parameters accurately.

PROBLEM DEFINITION

- Accurate wave energy forecasting is difficult due to complex and non-linear ocean waves
- Traditional physical and mathematical models have limited accuracy
- Prediction performance reduces during changing and extreme weather conditions
- There is a need for a reliable machine learning-based system
- The system should accurately predict wave power density.



SOLUTION

Use **Machine Learning models** to forecast wave energy accurately.

Apply models like **Random Forest, SVR, and Deep Learning (LSTM)** to learn patterns from historical wave and weather data.

Replace slow physical models with **data-driven prediction** for faster results.

Enable **real-time forecasting** using continuously updated sensor (buoy) data.

Improve energy harvesting efficiency and **smooth integration with the power grid**.

EXISTING SYSTEM

The existing system uses **traditional mathematical and physical models**.

It predicts wave energy based on **ocean wave theory**.

Data is collected from **wave buoys.**

Models like **Simulating Waves Nearshore** and **WAVEWATCH III** are used.

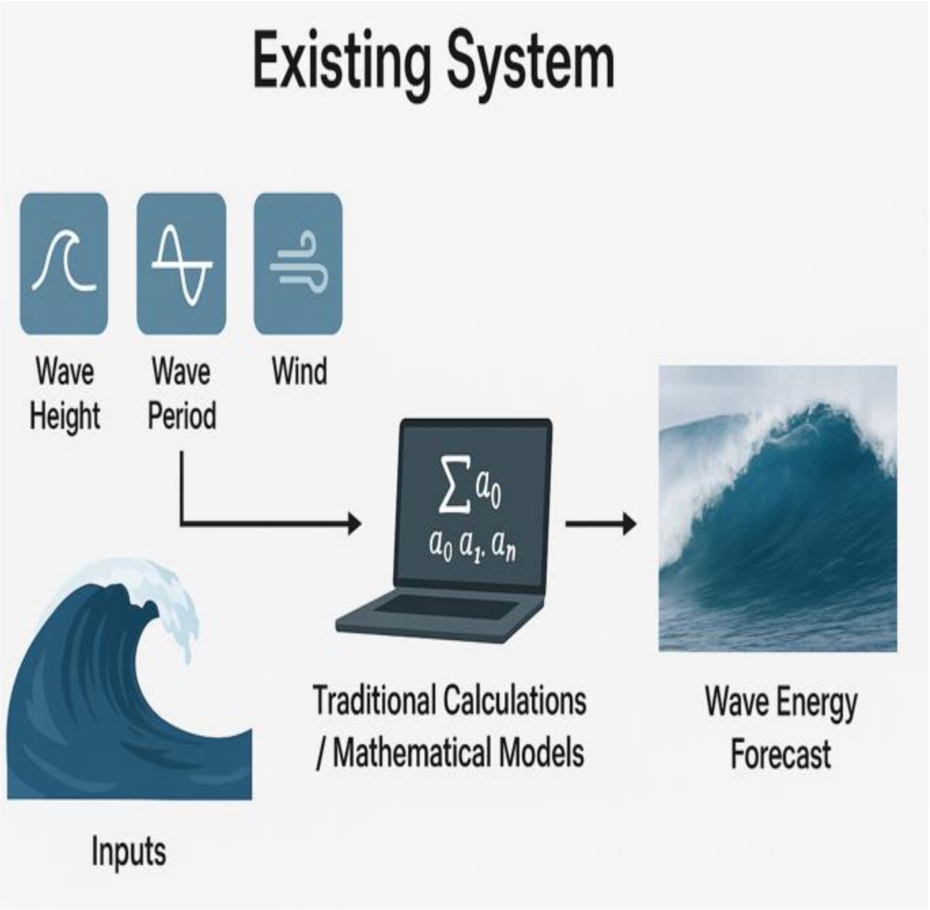
The system follows **fixed equations** to give predictions.

Input:

Wind speed and direction, sea surface conditions, initial wave parameters

Output:

Predicted wave height, wave period, estimated wave energy



PROPOSED SYSTEM

The proposed system uses **machine learning techniques**.

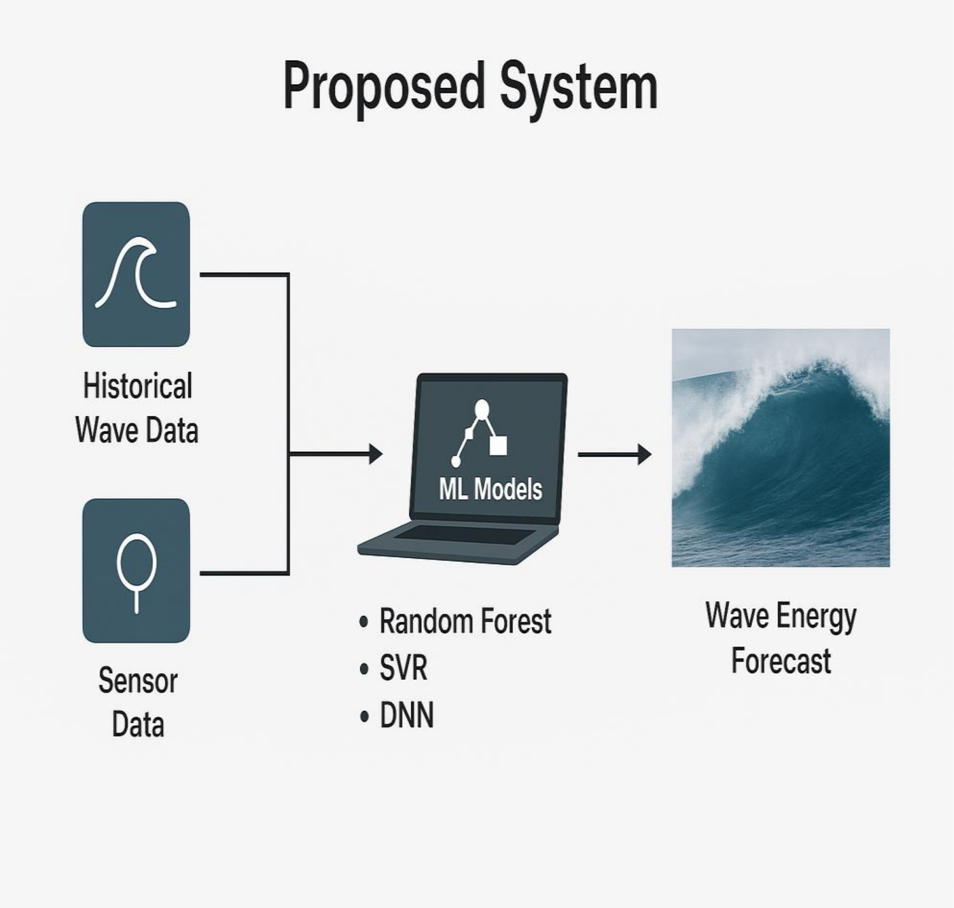
It learns patterns from **historical wave and weather data**.

Data is taken from **wave buoy datasets** .

Algorithms like **Random Forest, SVR, and Deep Learning** are used.

The system can handle **non-linear and complex wave behavior**.

Accuracy: Evaluated using RMSE,MAE,R^2 $score$



ADVANTAGES & DISADVANTAGES

ADVANTAGES(proposed System)

- Enhanced forecasting accuracy through data-driven learning.
- Reduced reliance on manual calibration and expert input.
- Real-time adaptability and faster computation.
- Scalable integration with IoT and big data platforms.

DISADVANTAGES(Existing System)

- High computational cost and time consumption for model execution.
- Poor adaptability to real-time environmental changes.
- Dependency on domain expertise for calibration and tuning.
- Limited scalability and performance with noisy or missing data.

SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS:

System	:	Intel i7
Hard Disk	:	1 TB.
Monitor	:	14’ Colour Monitor.
Mouse	:	Optical Mouse.
Ram	:	8GB.

SOFTWARE REQUIREMENTS:

Operating system	:	Windows 10.
Coding Language	:	Python.
Front-End	:	Html. CSS
Designing	:	Html, CSS, JavaScript.
Data Base	:	SQLite

ALGORITHMS

1 Random Forest

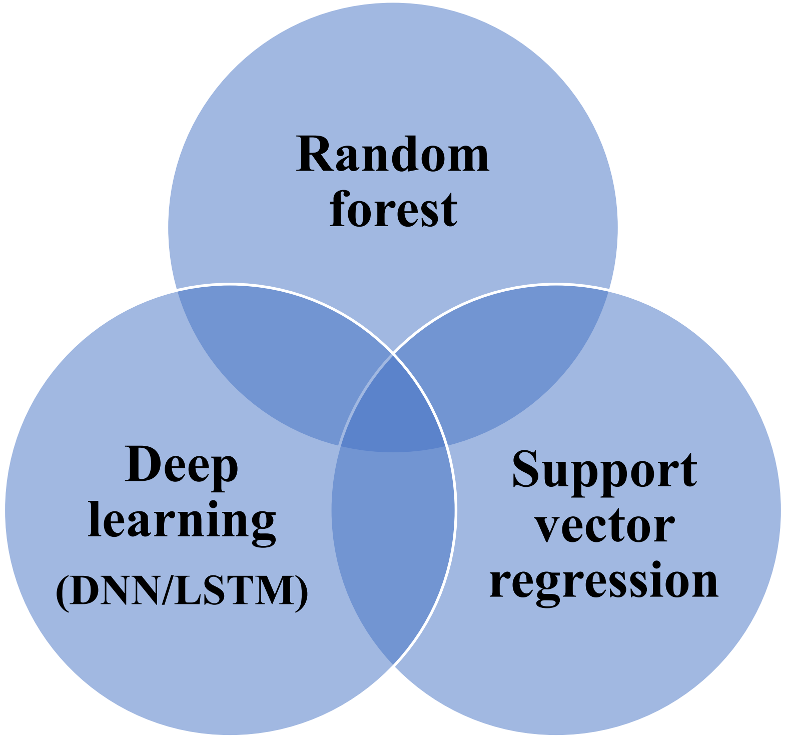
Uses many **decision trees** together
Works well with **non-linear data**
Gives **high accuracy**

2 Support Vector Regression (SVR)

Used to predict **continuous values**
Finds the best line or curve for prediction
Suitable for **wave height** and **energy prediction**

3 Deep Learning (DNN / LSTM)

Uses multiple layers to learn **complex patterns**
Works well with **large datasets**
Suitable for **time-series wave data**



MODULES

1. User Management Module
•Allows users to register using valid email and mobile number.
•Provides secure login access after admin approval.

2. Admin Management Module
•Activates registered users and controls system access.
•Monitors users, datasets, and forecasting results.

3. Dataset Upload & Management Module
•Supports uploading wave energy datasets in predefined format.
•Manages training and testing data for model execution.

4. Data Preprocessing Module
•Removes missing values, noise, and inconsistent data.
•Normalizes data to improve machine learning performance.

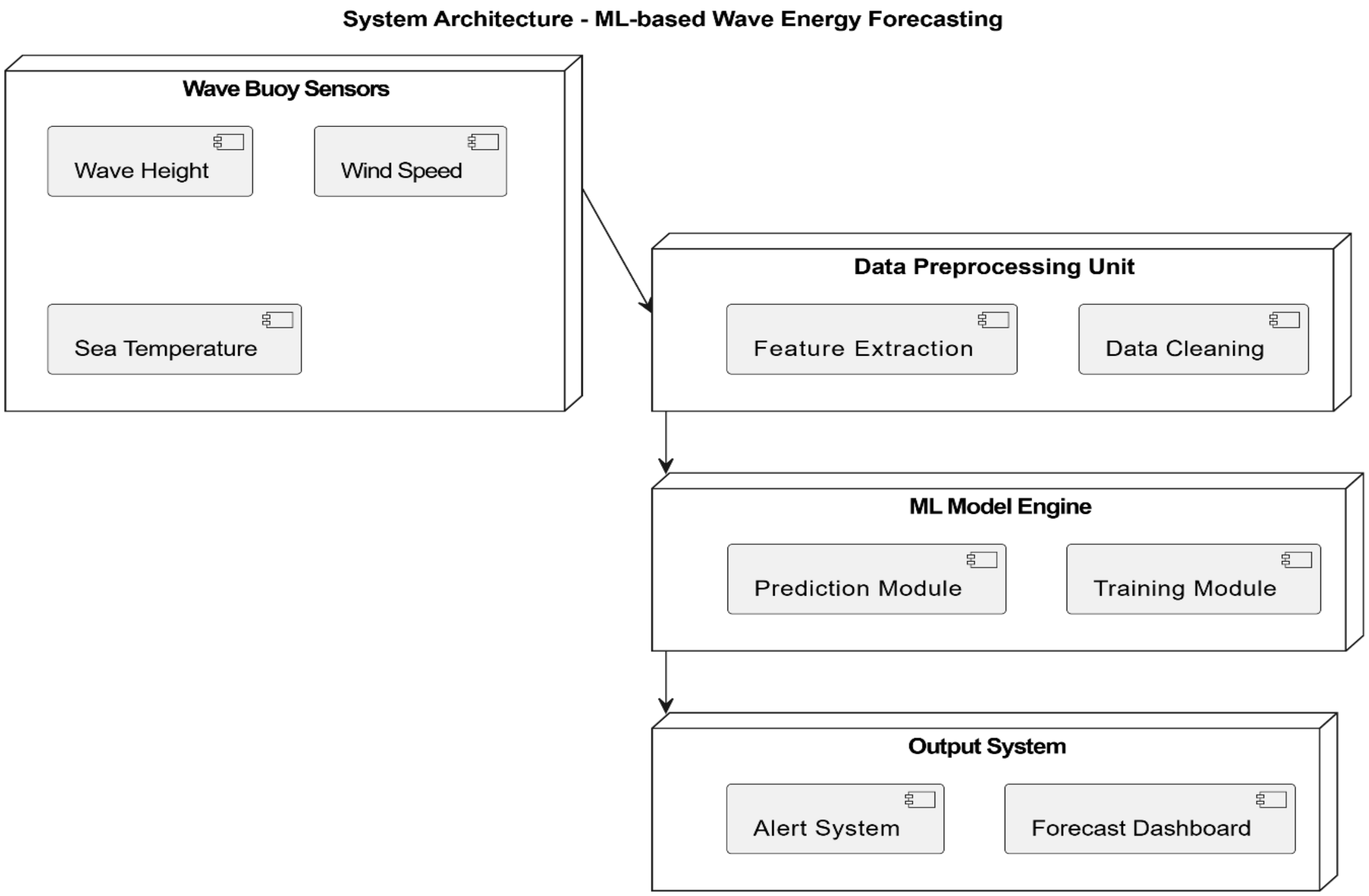
5. Feature Selection Module
•Identifies important wave parameters affecting energy prediction.
•Reduces computational complexity and improves accuracy.

6. Machine Learning Model Module
•Trains Random Forest, SVR, and Deep Learning models.
•Learns non-linear and time-series wave patterns.

7. Prediction & Forecasting Module
•Predicts wave energy and wave parameters accurately.
•Supports efficient renewable energy generation planning.

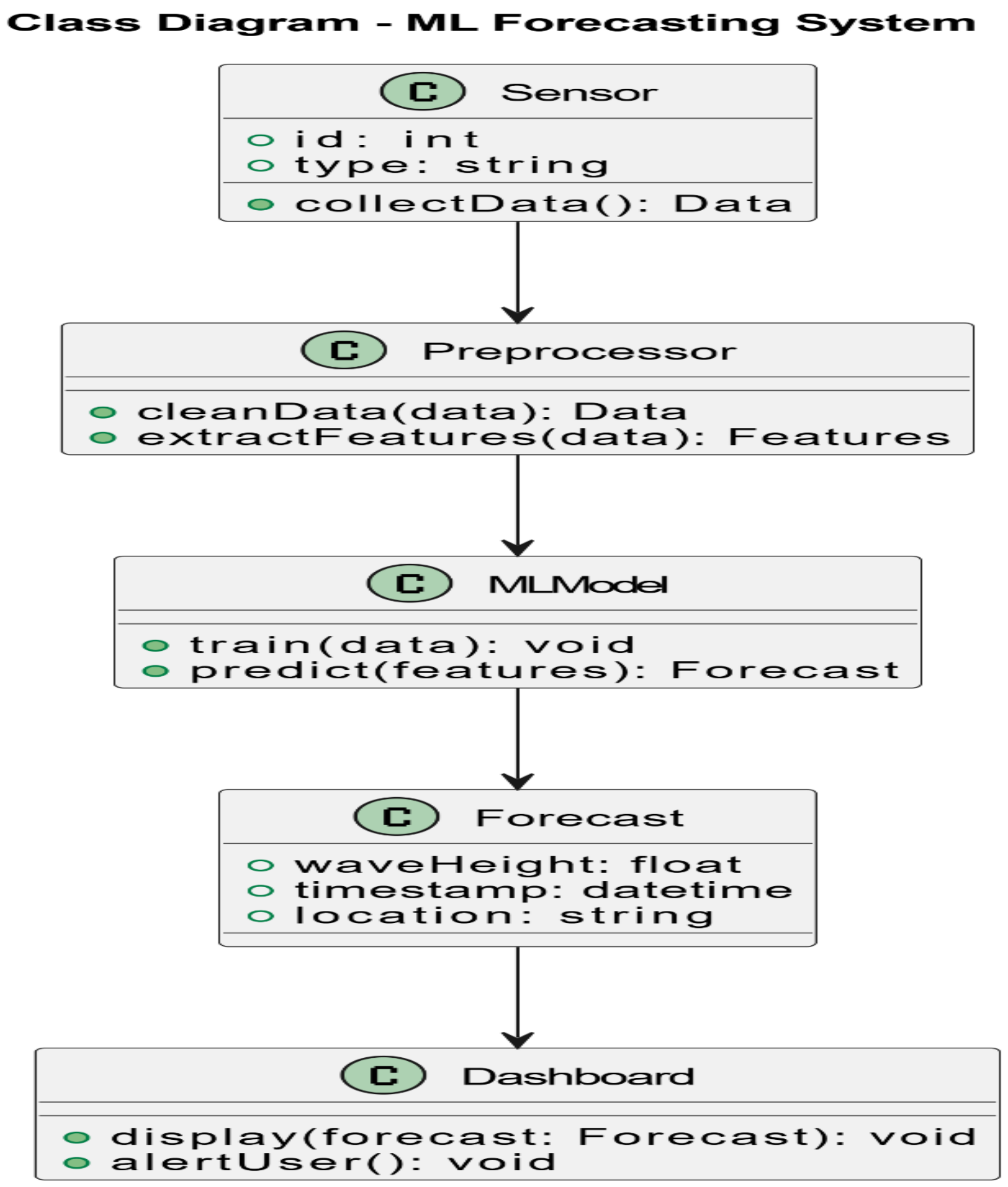
8. Visualization & Results Module
•Displays prediction results using graphs and charts.
•Helps in easy analysis and performance evaluation.

SYSTEM ARCHITECTURE

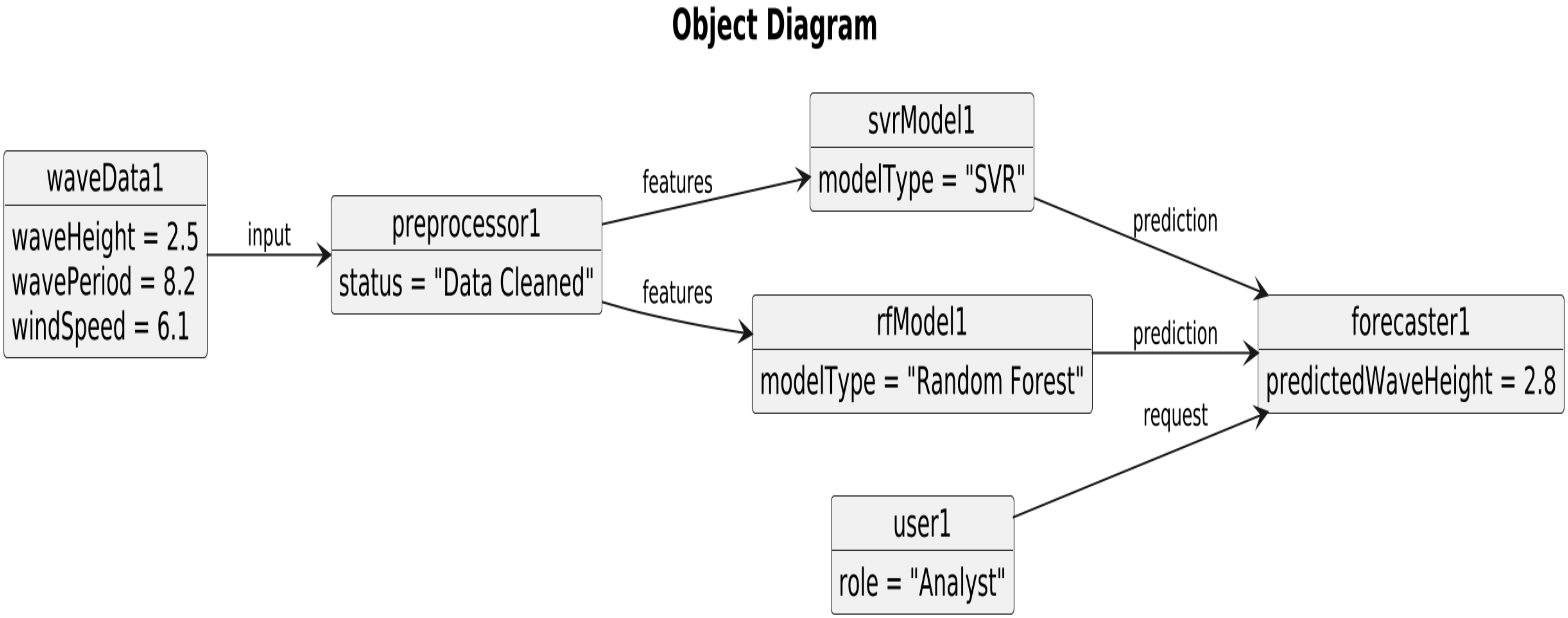


UML DIAGRAMS :

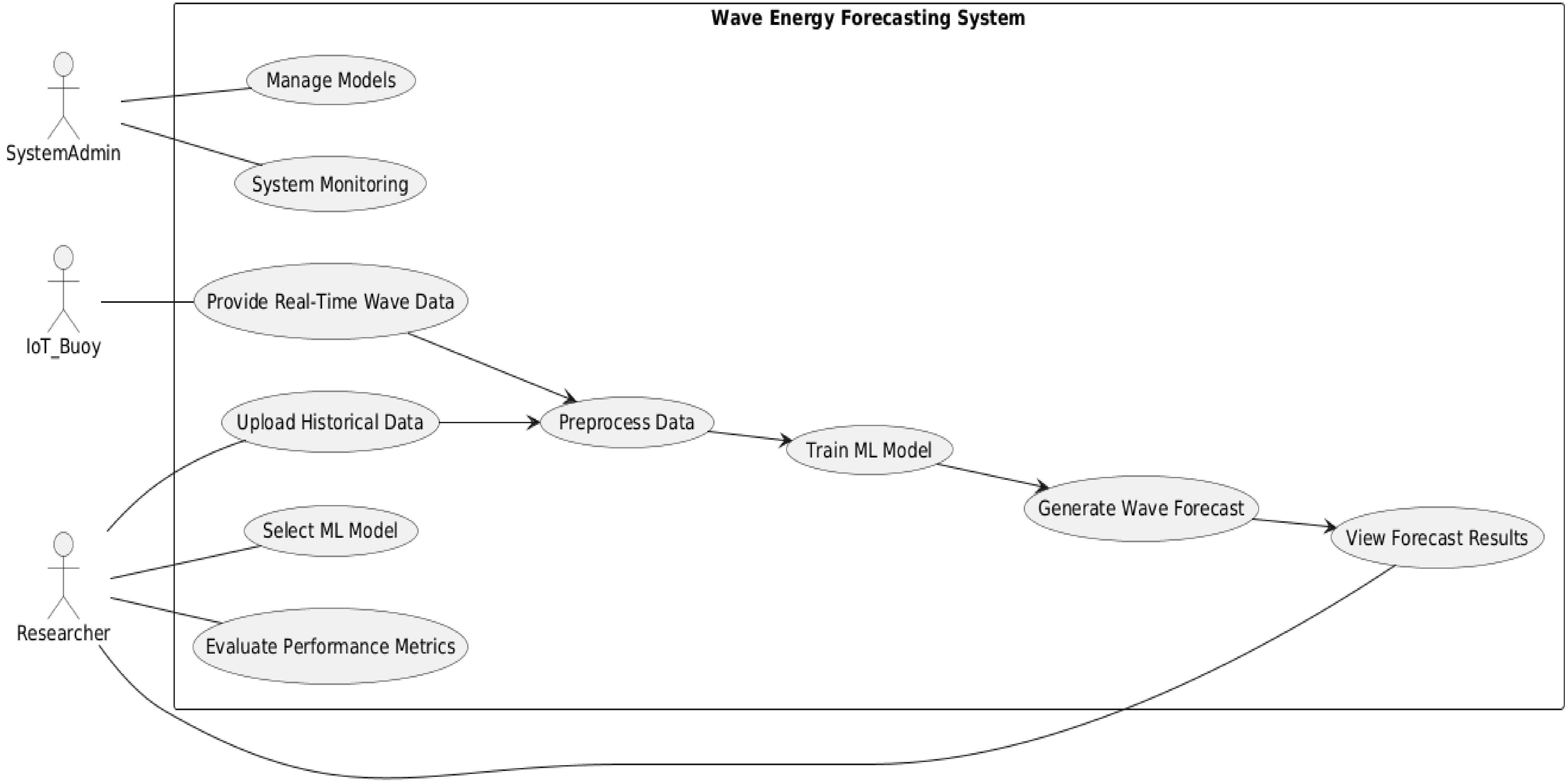
CLASS DIAGRAM:



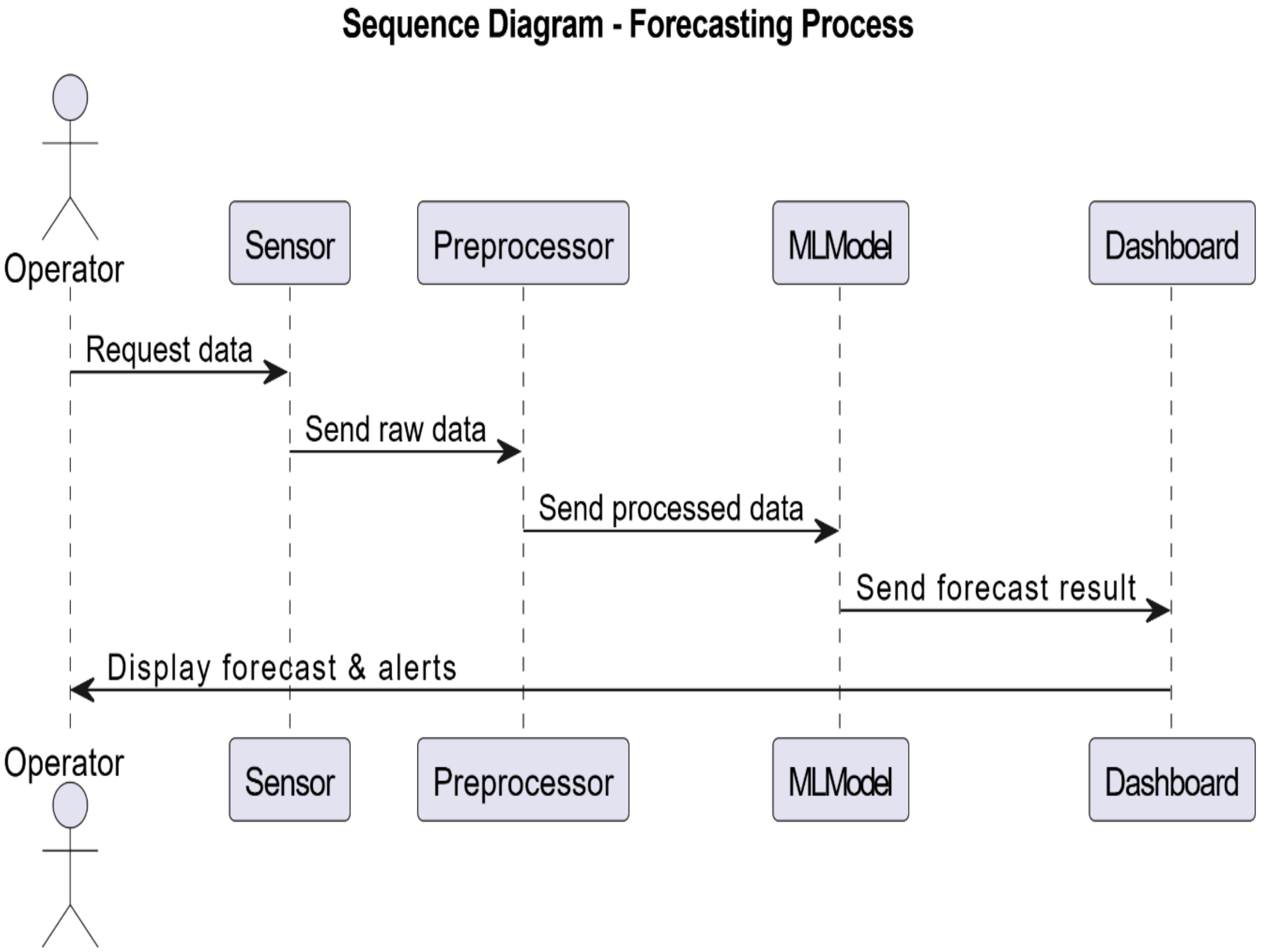
OBJECT DIAGRAM:



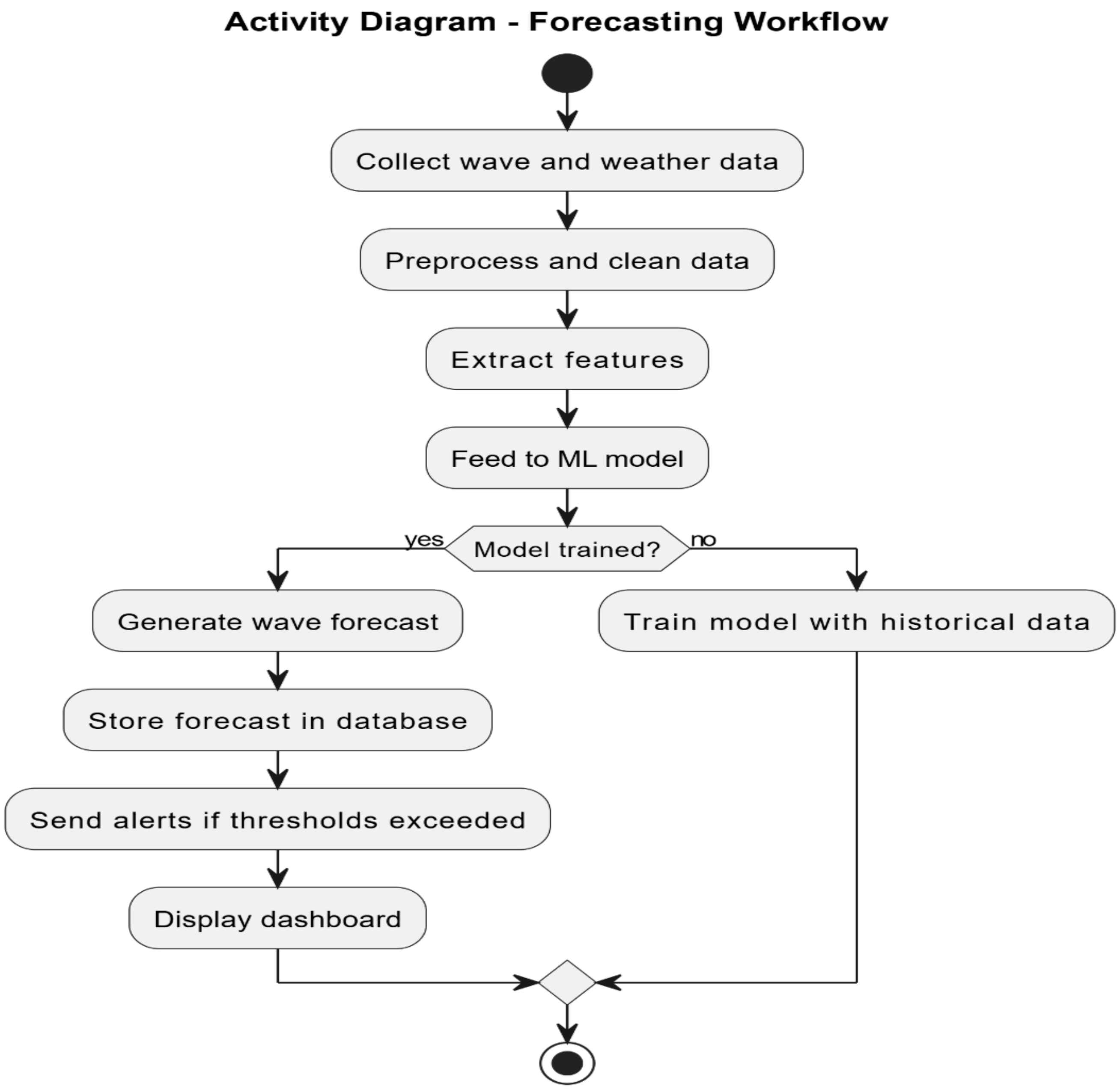
USE CASE DIAGRAM:



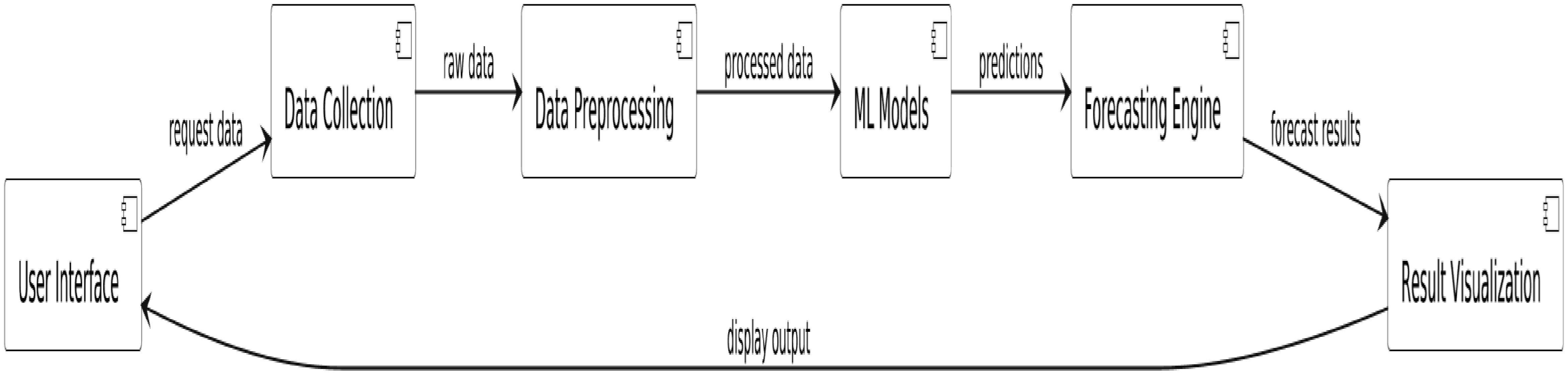
SEQUENCE DIAGRAM:



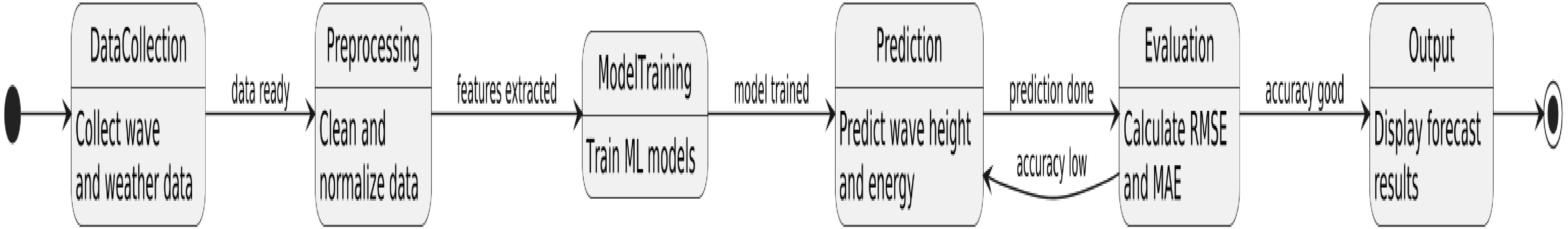
ACTIVITY DIAGRAM:



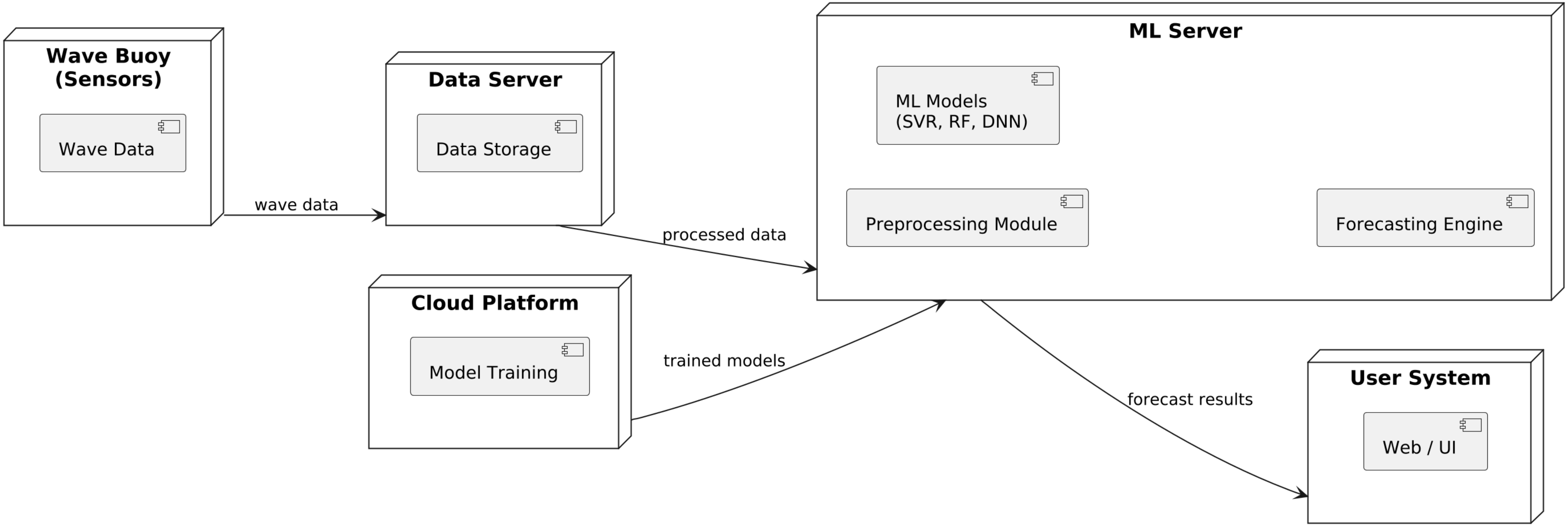
COMPONENT DIAGRAM:



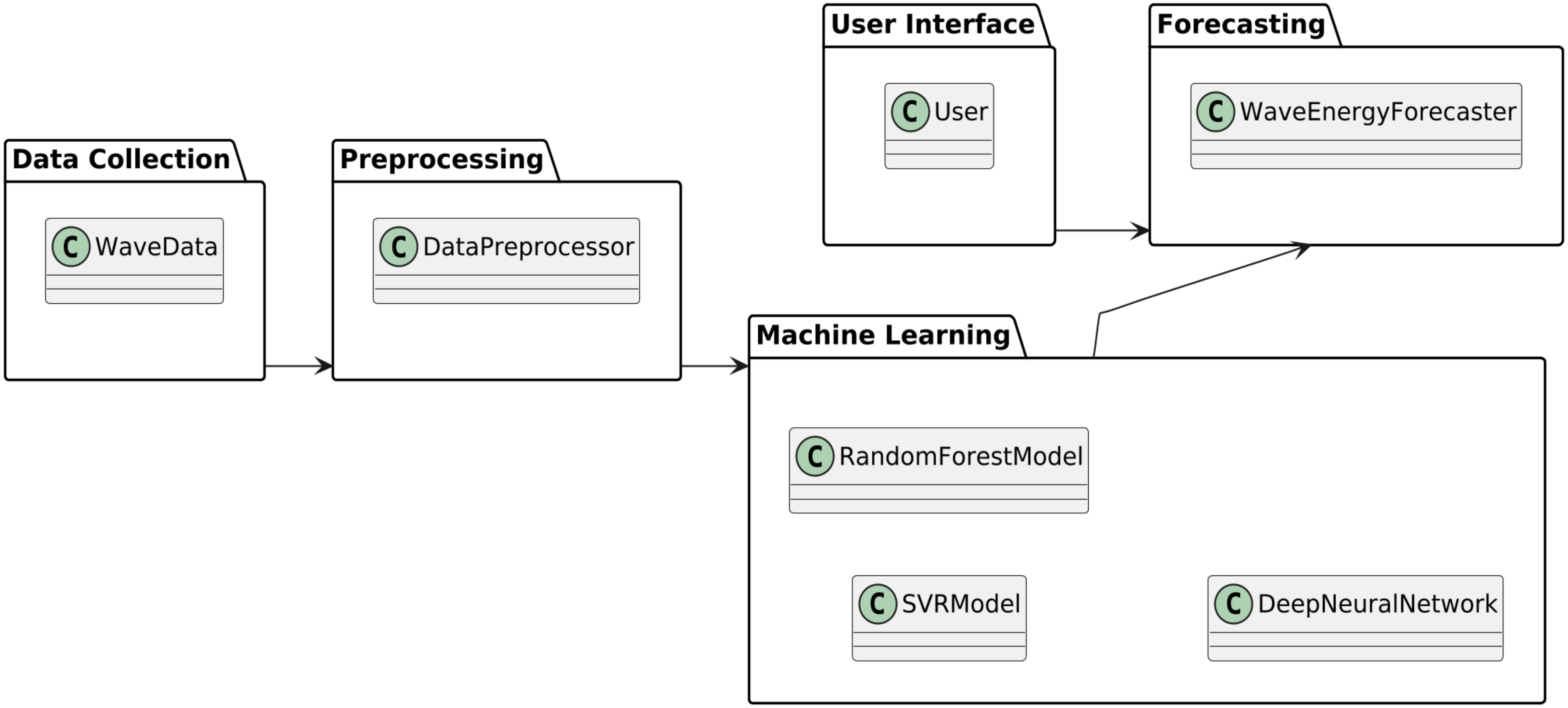
STATE MACHINE DIAGRAM:



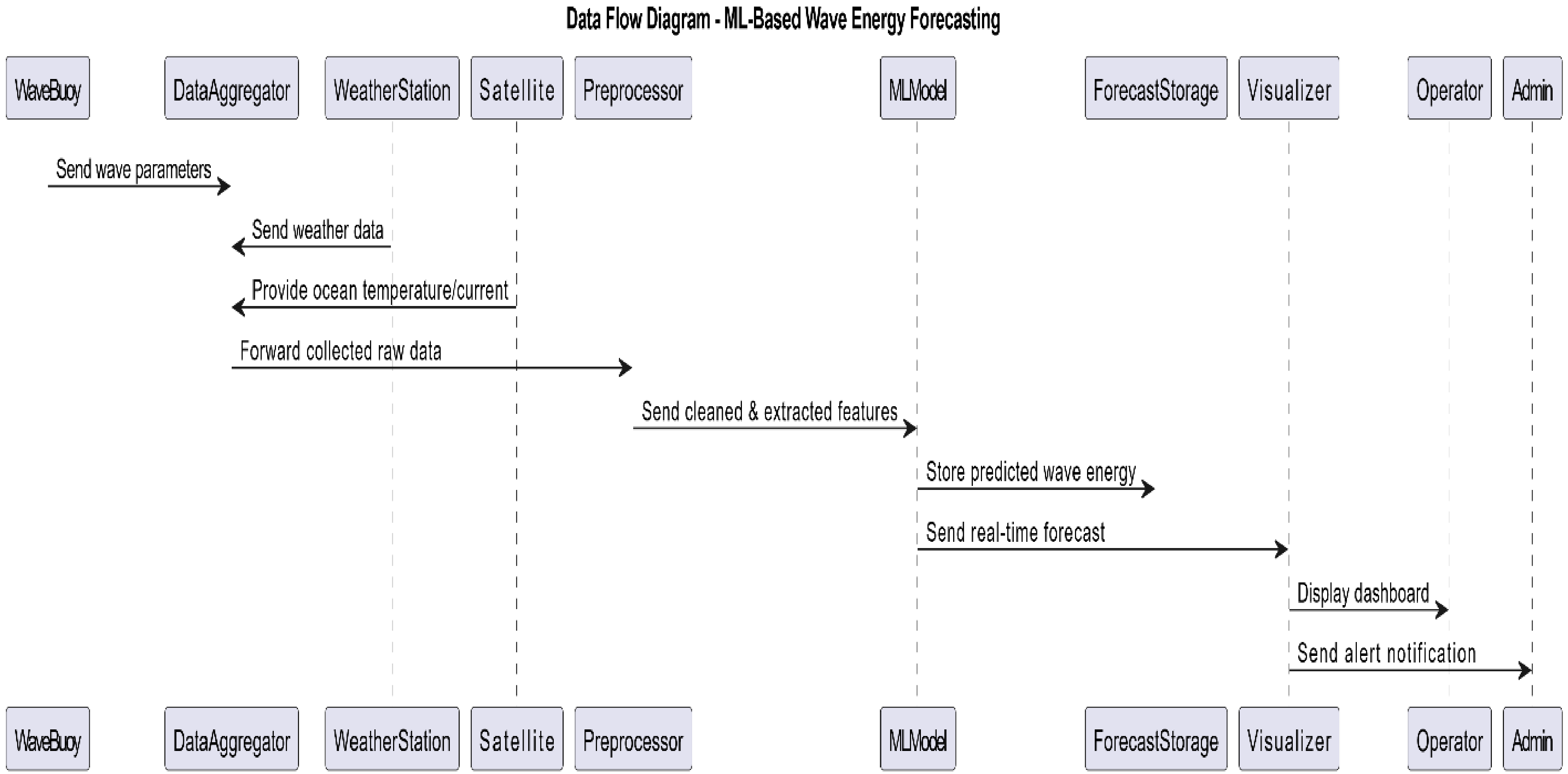
DEPLOYMENT DIAGRAM:



PACKAGE DIAGRAM:

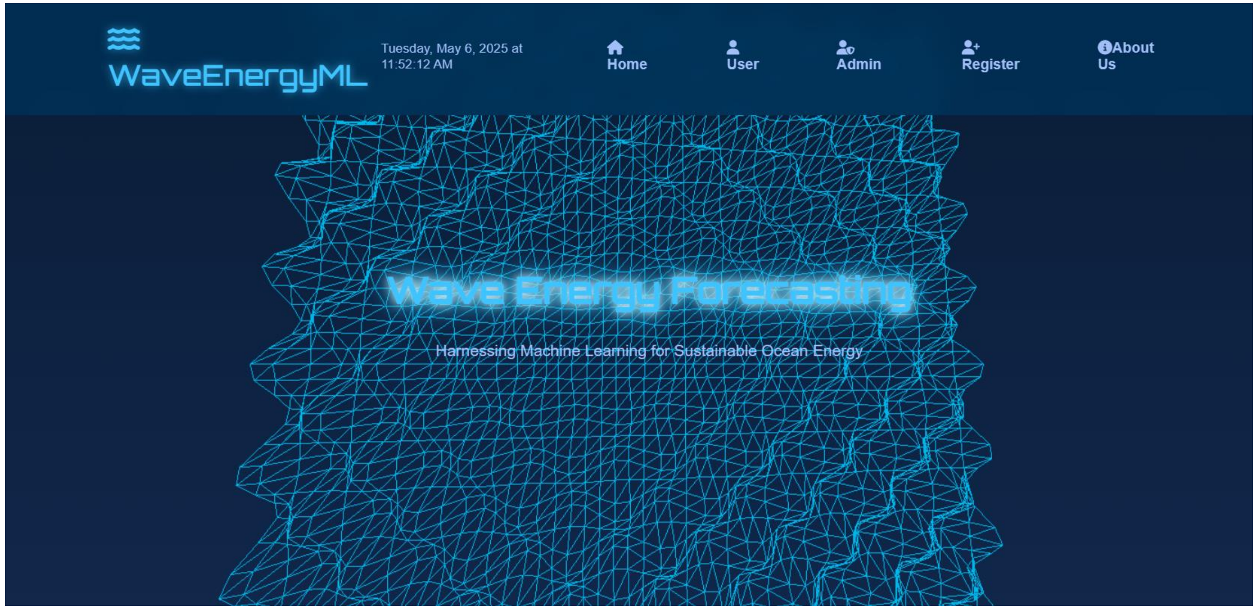


DATA FLOW DIAGRAM:

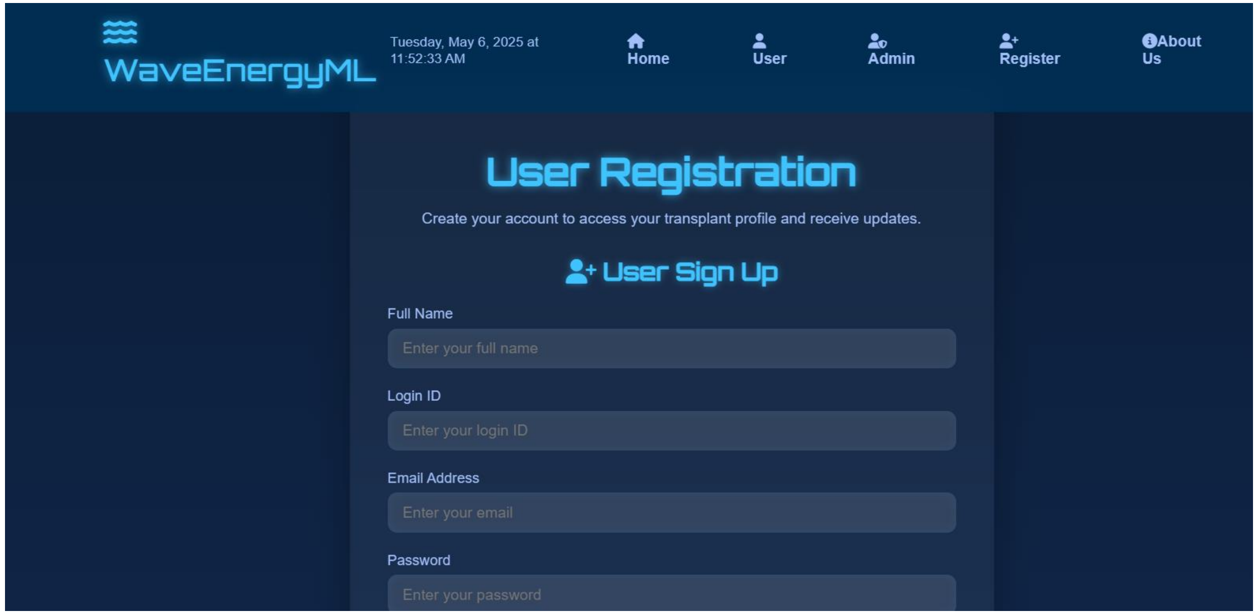


SCREENSHOTS:

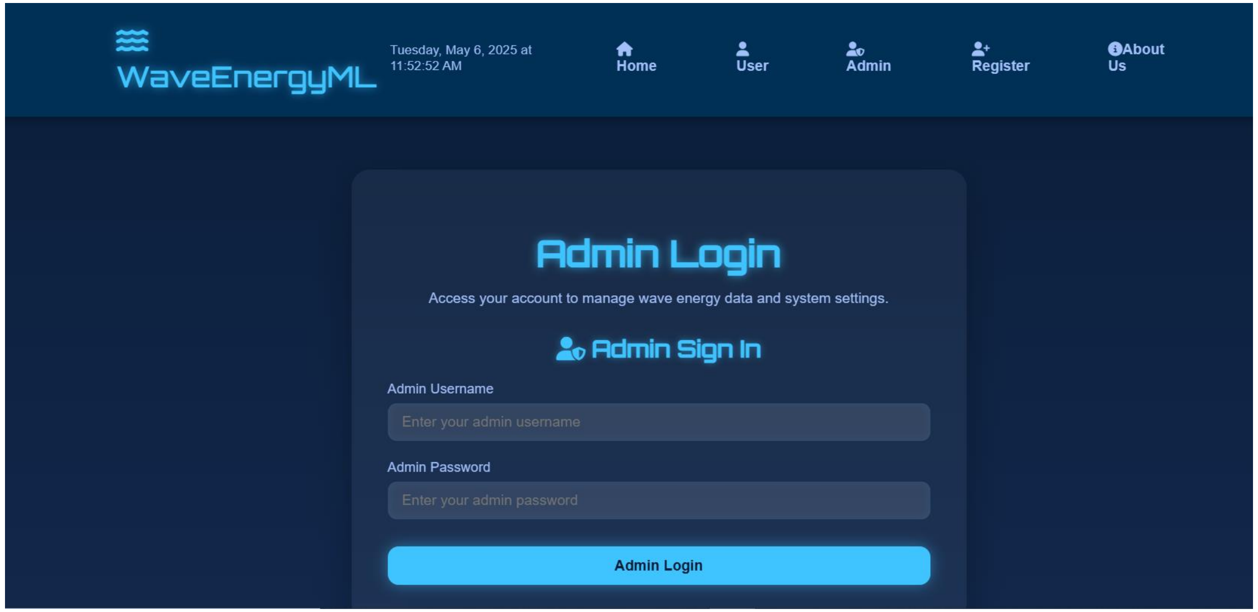
Home Page:



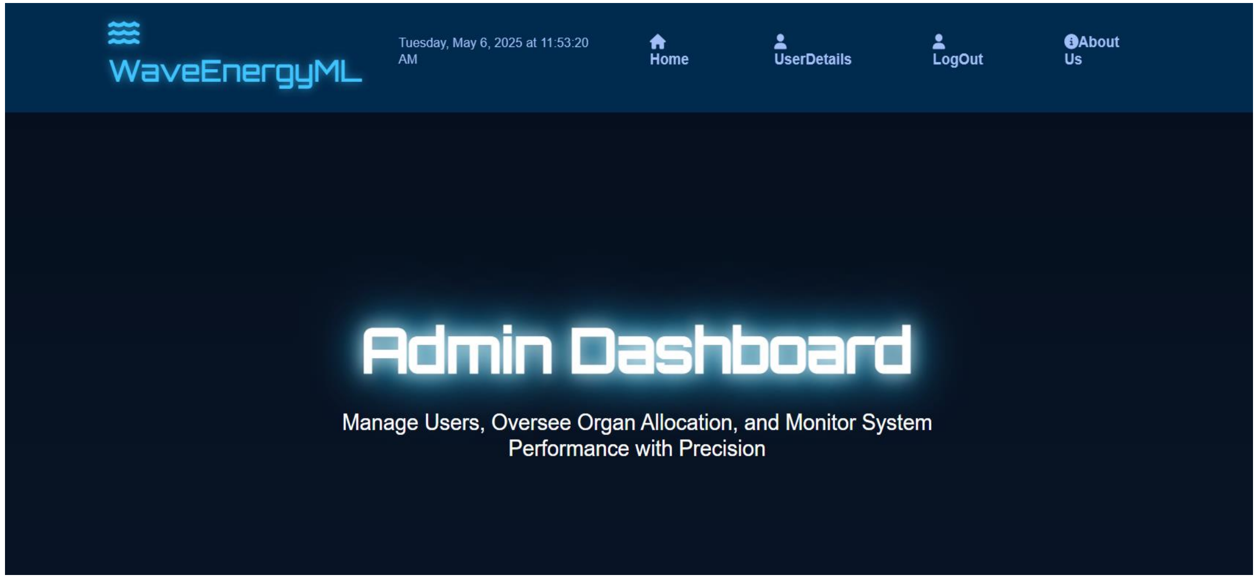
User Register Page:



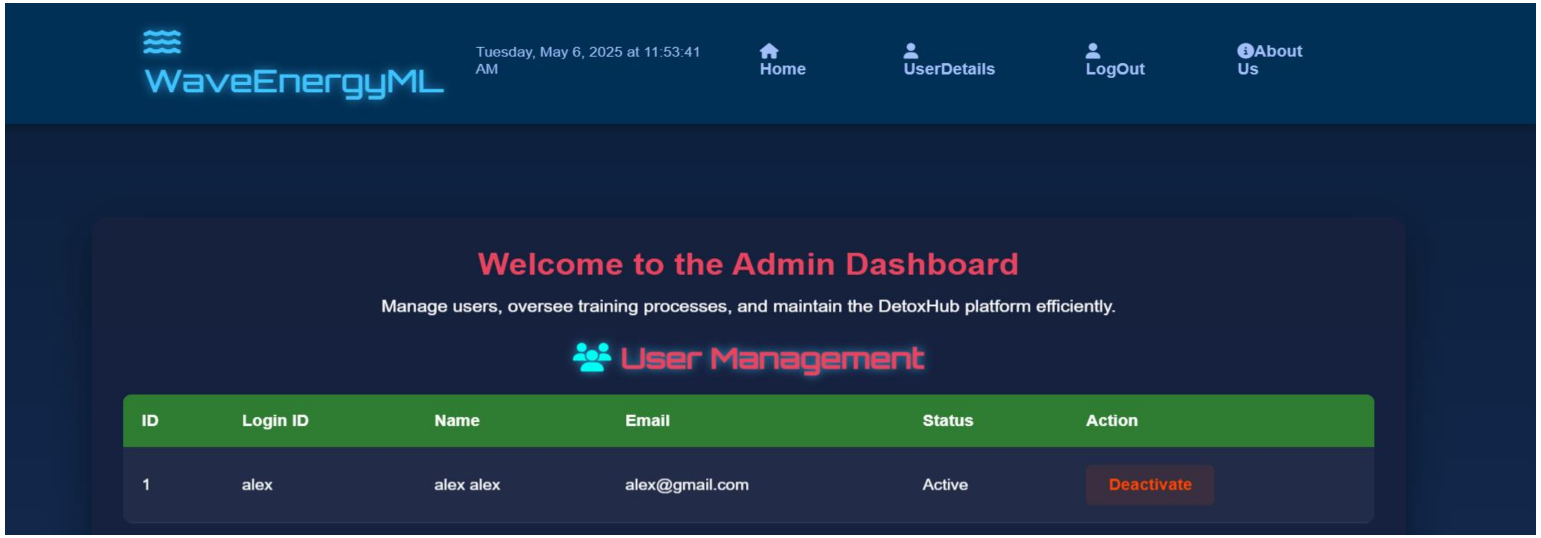
Admin Login Page:



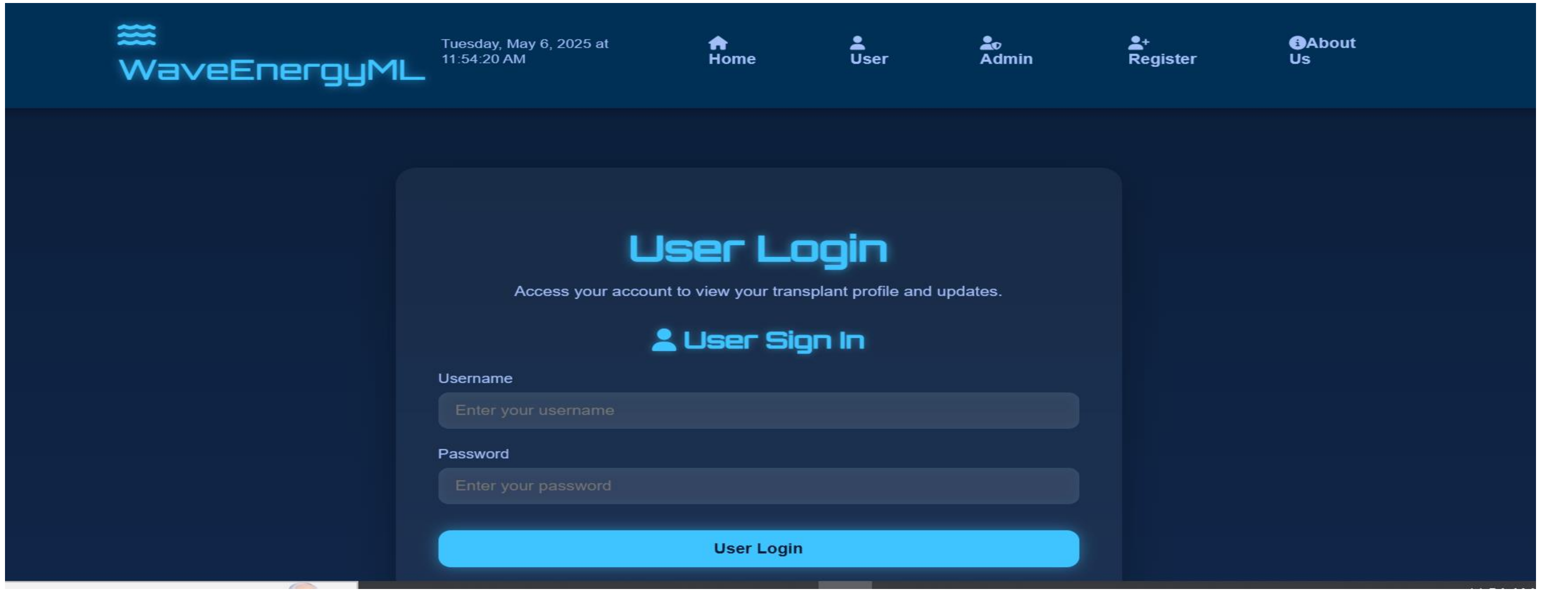
Admin Home Page:



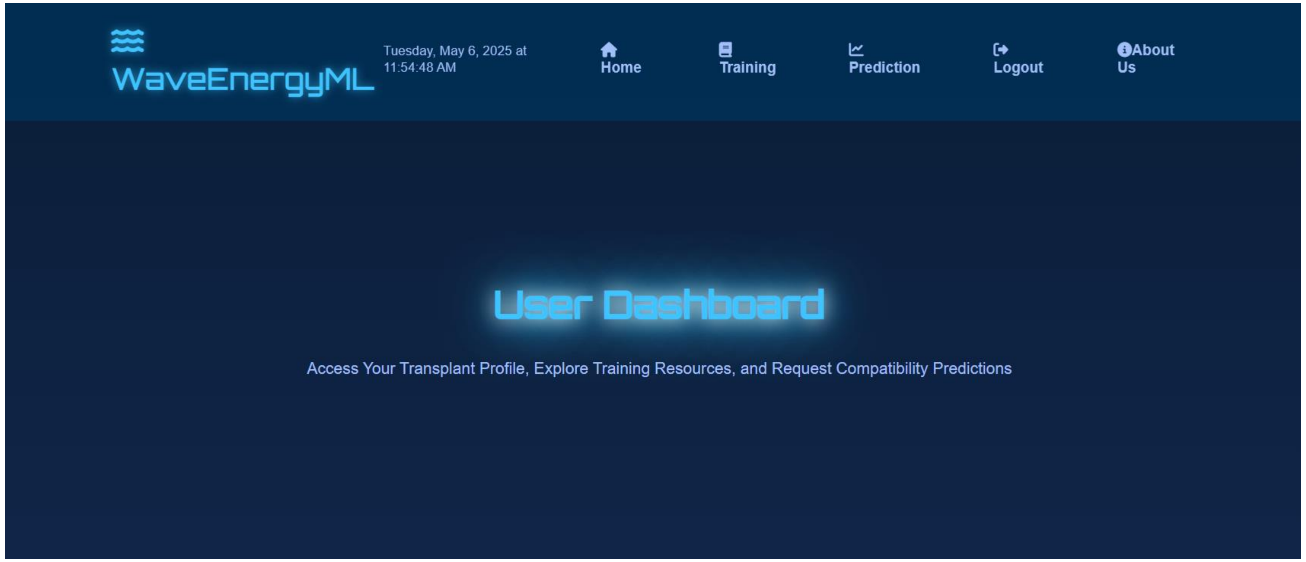
User Activation Page:



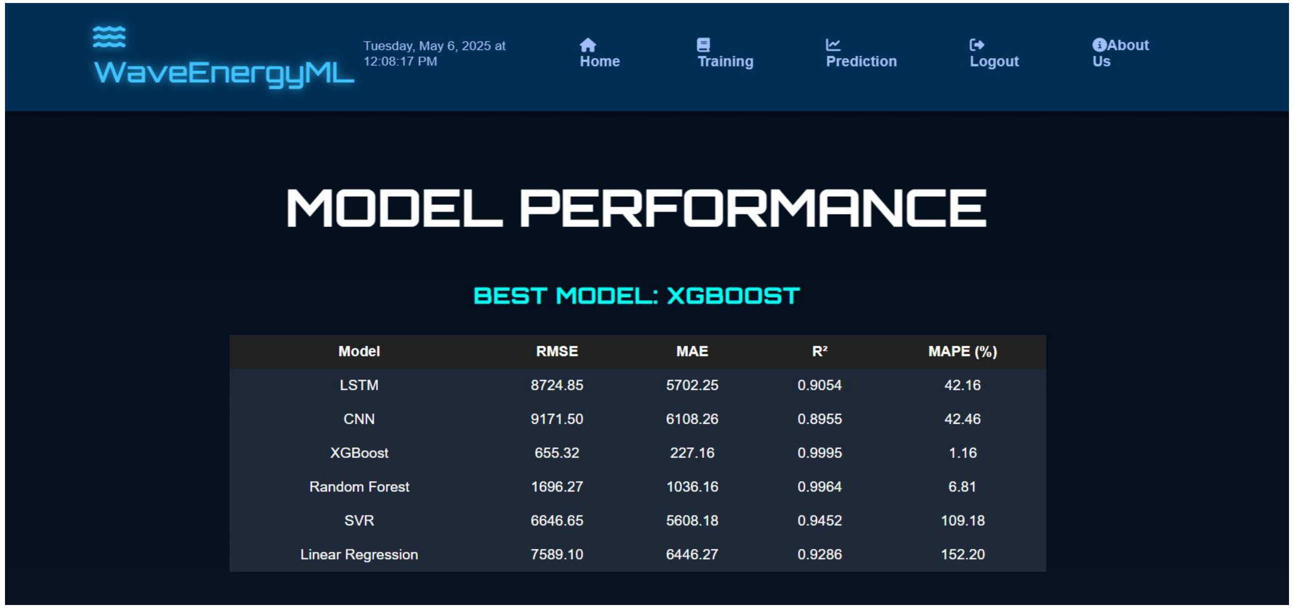
User Login Page:



User Home Page:

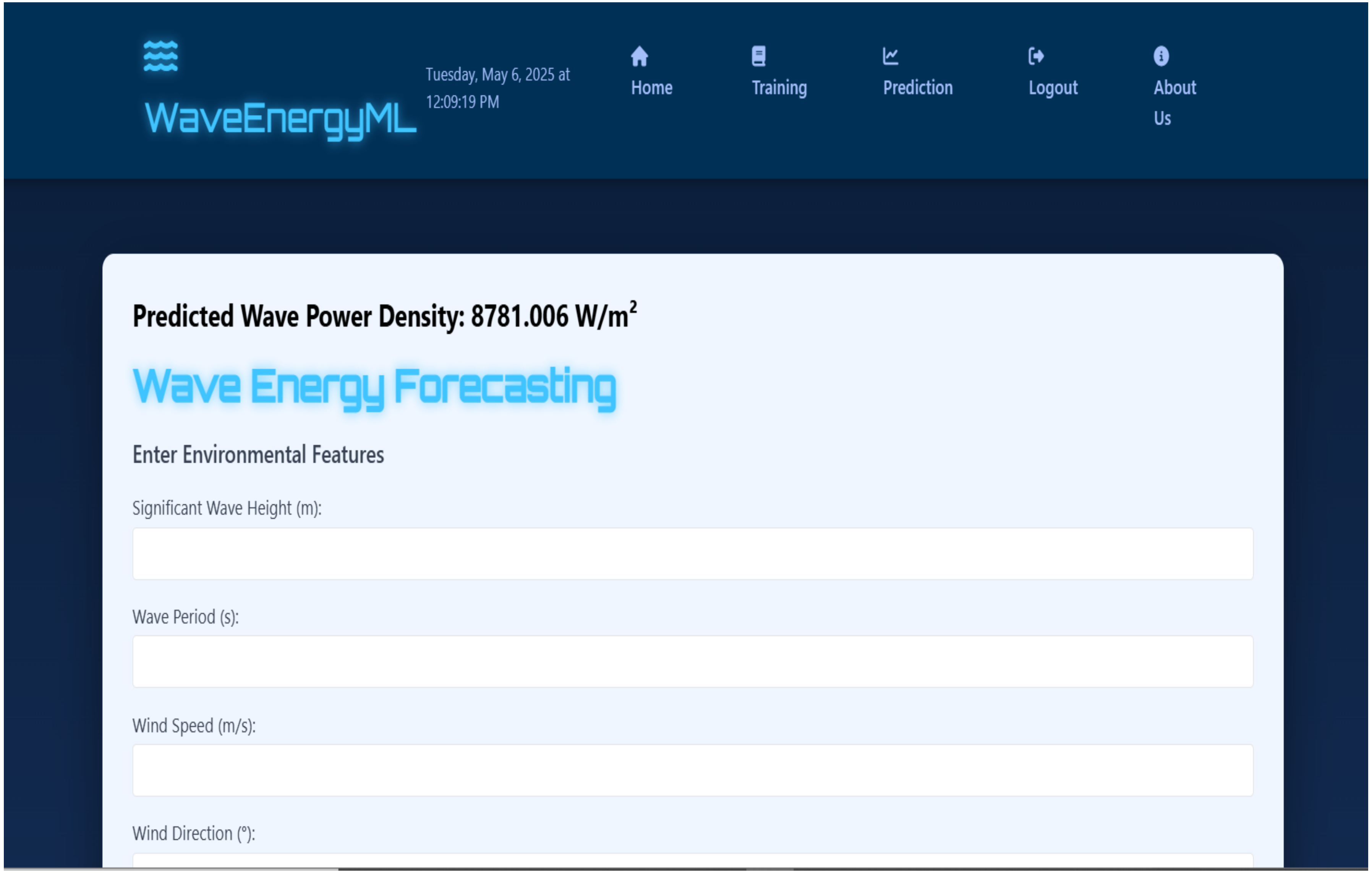


Training Page:



Model	RMSE	MAE	R^2	MAPE (%)
LSTM	8724.85	5702.25	0.9054	42.16
CNN	9171.50	6108.26	0.8955	42.46
XGBoost	655.32	227.16	0.9995	1.16
Random Forest	1696.27	1036.16	0.9964	6.81
SVR	6646.65	5608.18	0.9452	109.18
Linear Regression	7589.10	6446.27	0.9286	152.20

Prediction Page:



CONCLUSION

- This project shows that machine learning models can accurately forecast wave energy by learning patterns from wave and weather data.
- Compared to traditional methods, the proposed system handles complex and changing ocean conditions more effectively
- As a result, it supports efficient renewable energy generation and smooth integration into the power grid.

REFERENCES

J. Yan, N. Mei, D. Zhang, Y. Zhong, and C. Wang, “Review of wave power system development and research on triboelectric nano power systems,” Frontiers in Energy Research, vol. 10, Jul. 2022, Doi: 10.3389/fenrg.2022.966567.

O. Farrok, K. Ahmed, A. D. Tahlil, M. M. Farah, M. R. Kiran, and Md. R. Islam, “Electrical Power Generation from the Oceanic Wave for Sustainable Advancement in Renewable Energy Technologies,” Sustainability, vol. 12, no. 6, p. 2178, Mar. 2020, Doi: 10.3390/su12062178.

C. Zheng et al., “An assessment of global ocean wave energy resources over the last 45 a,” Acta Oceanological Sinica, vol. 33, no. 1, pp. 92–101, Jan. 2014, Doi: 10.1007/s13131-014-0418-5.

X. Zhang, Y. Li, S. Gao, and P. Ren, “Ocean Wave Height Series Prediction with Numerical Long Short-Term Memory,” Journal of Marine Science and Engineering, vol. 9, no. 5, p. 514, May 2021, DOI: 10.3390/jmse9050514.

A. S. Bastos, T. R. C. De Souza, D. S. Ribeiro, M. De Lourdes Noronha Motta Melo, and C. B. Martinez, “Wave energy Generation in Brazil: A Georeferenced oscillating Water column inventory,” Energies, vol. 16, no. 8, p. 3409, Apr. 2023, DOI: 10.3390/en16083409.

S. Vitousek and O. B. Fringer, “Physical vs. numerical dispersion in nonhydrostatic ocean modeling,” Ocean Modelling, vol. 40, no. 1, pp.

72–86, Jan. 2011, DOI: 10.1016/j.ocemod.2011.07.002

M. A. Benitz, M. A. Lackner, and D. P. Schmidt, “Hydrodynamics of offshore structures with specific focus on wind energy applications,” Renewable and Sustainable Energy Reviews, vol. 44, pp. 692–716, Jan. 2015, DOI: 10.1016/j.rser.2015.01.021.

A. Shadmani, M. R. Nikoo, A. H. Gandomi, R.-Q. Wang, and B. Golparvar, “A review of machine learning and deep learning applications in wave energy forecasting and WEC optimization,” Energy Strategy Reviews, vol. 49, p. 101180, Sep. 2023, DOI: 10.1016/j.esr.2023.101180.

T. Breunung and B. Balachandran, “Data-driven, high resolution ocean wave forecasting and extreme wave predictions,” Ocean Engineering, vol. 268, p. 113271, Dec. 2022, DOI: 10.1016/j.oceaneng.2022.113271.

Thank You